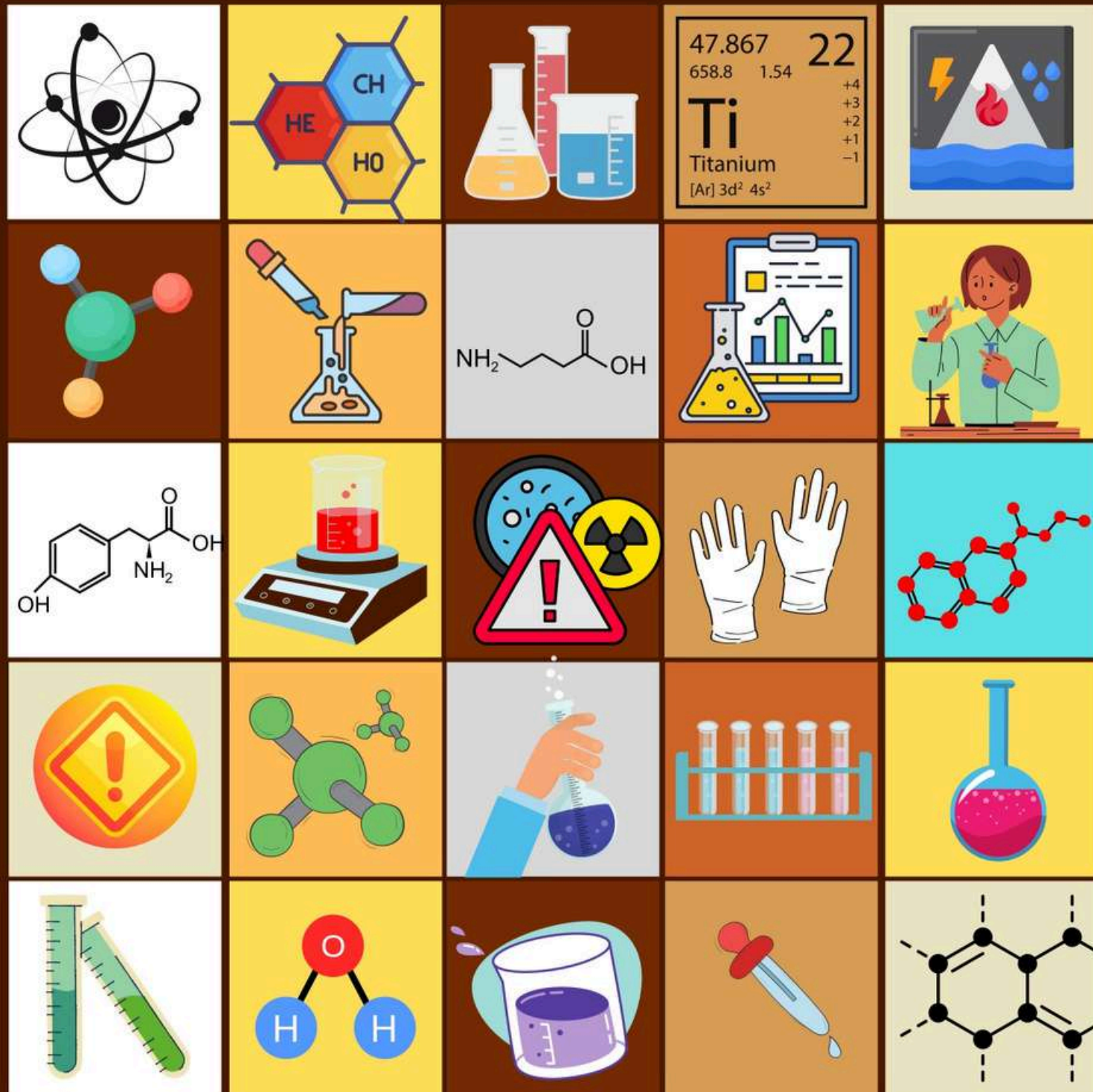




CLASS 9th NOTES CHEMSITRY

IS MATTER AROUND US PURE?

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IS MATTER AROUND US PURE?



Substance :

A substance is a kind of matter that cannot be separated into other kind of matter by any physical process.

Pure Substance :

A pure substance is a form of matter that has a definite composition and distinct properties. They are made of only one entity.

e.g. Milk, water, diamond, etc.



Mixture :

Mixtures are combinations of different substances, either elements or compounds, which retain their individual properties.

e.g. Crude Oil, Seawater, etc.

Types of mixtures :-

Homogeneous mixture : A mixture which has a uniform composition throughout is called a homogeneous mixture or solution.

Examples: sugar in water, salt in water.

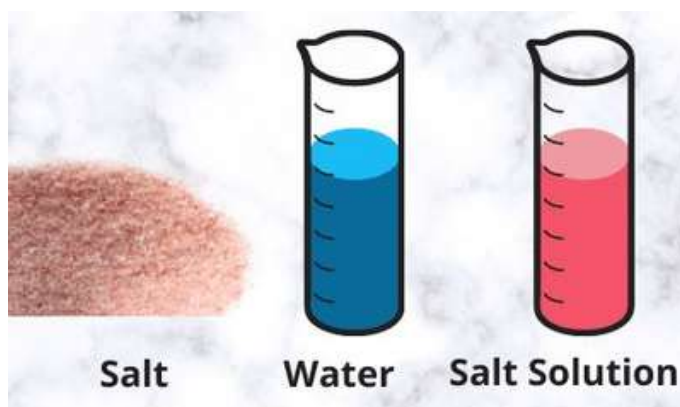
Heterogeneous mixture : A mixture which contains physically distinct parts and has a non-uniform composition is called a heterogeneous mixture.

Examples: Mixture of salt and iron filings, sand and sugar.

Solution : A solution is a homogeneous mixture of two or more substances. e.g. Soda water, Lemonade, etc.

Components of a Solution:

- **Solute:** The substance that is dissolved in the solvent. It is present in a lesser amount.
- **Solvent:** The substance in which the solute is dissolved. It is present in a greater amount.



Alloys are mixtures of metals or a metal and a non-metal, unseparable by physical methods. Though not separable, they retain properties of constituents and can vary in compositions that's why they are considered as a mixture. Example: brass is 30% zinc and 70% copper.

Properties of a solution:

1. A solution is a **homogeneous** mixture.
2. The particles of a solution are **smaller** than **1 nm** (10^{-9} metre) in diameter. So, they cannot be seen by naked eyes.
3. Because of very small particle size, they **do not scatter a beam of light** passing through the solution. So, the path of light is not visible in a solution.
4. The **solute particles cannot be separated** from the mixture by the process of filtration. The solute particles do not settle down when left undisturbed, that is, a solution is stable

Concentration of Solution :

Mass by mass percentage

$$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

Mass by volume percentage

$$= \frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$$

Volume by volume percentage of a solution

$$= \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

	SOLIDS	LIQUIDS	GASES
1	Size of solute particles is smallest. $< 10^{-9}\text{m}$	Size of solute particles bigger than true but smaller than suspension. In between 10^{-9} to 10^{-6}m .	Size of particles biggest. $> 10^{-6}\text{m}$
2	Solute particles can't be seen with naked eye.	Solute particles can't be seen with naked eye.	Can be seen with naked eye.
3	Homogeneous mixture.	Seems homogeneous but actually heterogeneous.	Heterogeneous mixture.
4	Particles can't be separated by filtration .	Particles can't be separated by filtration .	Can be separated by filtration.
5	Transparent	Translucent	Opaque
6	Stable solutions - i.e. solute particles do not settle on keeping.	Stable solutions	Unstable solution-solute particles settle upon keeping.
7	Do not show tyndall effect.	Show tyndall effect.	May or may not show tyndall effect.
8	Solutions diffuse rapidly through filter paper as well as parchment paper.	Colloid particles pass through filter paper but not through parchment paper.	Suspension particles do not pass through filter paper as well as parchment paper.
9	e.g. Sugar in water.	e.g. Milk, blood.	e.g. Sand/mud in water.

Suspension: A suspension is a heterogeneous mixture in which the solute particles do not dissolve but remain suspended throughout the bulk of the medium basically suspension is non - homogeneous systems in which solids are dispersed in liquids.

Properties of a suspension:

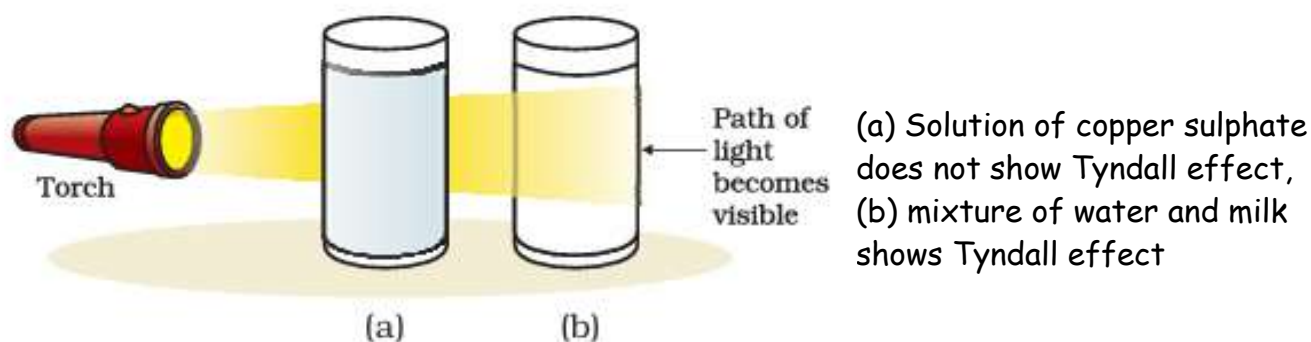
- Suspension is a **heterogeneous** mixture.
- Particles are **visible** to the naked eye.

- Particles scatter light, making the **path visible**.
- Solute particles settle over time, making the **suspension unstable**.
- Particles can be separated by **filtration**.
- When particles settle, the suspension breaks and no longer scatters light.

Colloidal Solution

The mixture in which one substance of dispersed insoluble particles is suspended throughout the other substance is known as colloidal solution or colloids.

The phenomenon of scattering of a beam of light by colloidal particles, when passed through a colloidal solution, is called the **Tyndall effect**.



Properties of Colloidal solution:

1. A colloid is a **heterogeneous** mixture.
2. Particle size is **too small** to be seen by naked eyes.
3. Particles scatter light, making its **path visible**.
4. Colloids are stable and **do not settle** when left undisturbed.
5. Cannot be separated by filtration; requires centrifugation.

Common examples of colloids :

Dispersed Phase : The solute-like component of the dispersed particles in a colloid form the dispersed phase.

Dispersion Medium : The component in which the dispersed phase is suspended is known as the dispersing medium.

Aerosol : A colloidal solution with dispersed phase solid/liquid and dispersing medium gas is called Aerosol. e.g. clouds.

Foam : A colloidal solution with dispersed phase gas and dispersing medium solid/liquid is called Foam. e.g. Shaving cream.

Dispersed phase	Dispersing Medium	Type	Example
Liquid	Gas	Aerosol	Fog, clouds, mist
Solid	Gas	Aerosol	Smoke, automobile exhaust
Gas	Liquid	Foam	Shaving cream
Liquid	Liquid	Emulsion	Milk, face cream
Solid	Liquid	Sol	Milk of magnesia, mud
Gas	Solid	Foam	Foam, rubber, sponge, pumice
Liquid	Solid	Gel	Jelly, cheese, butter
Solid	Solid	Solid Sol	Coloured gemstone, milky glass

Ye table bhi dekhlena



Physical and Chemical Changes:

Physical change	Chemical change
1. In a physical change, only physical properties such as colour, physical state, density, volume, etc. change; chemical properties remain unchanged. 2. No new substance is formed in a physical change. 3. Very little or no energy in the form of heat, light or sound is usually absorbed or given out in a physical change. 4. A physical change is a temporary change. 5. The original form of substance can be regained by simple physical methods. 6. A physical change is reversible.	1. In a chemical change, the chemical composition and chemical properties undergo a change. 2. A new substance is formed in a chemical change. 3. A chemical change is always accompanied by absorption or evolution of energy. 4. A chemical change is a permanent change. 5. Original substance cannot be obtained by simple physical methods. 6. A chemical change is irreversible.

e.g. ice, water, and water vapor

e.g. Oil burns in air, water extinguishes fire.

Types of Pure Substances:

1 Elements - Basic form of matter that cannot be broken down by chemical reactions.

- **Categories:** Metals, non-metals, metalloids.

Property	Metals	Non-Metals
Lustre	Shiny (e.g., gold, silver)	Not shiny (e.g., carbon, sulfur)
Color	Silvery-grey or golden-yellow	Various colors (e.g., oxygen - colorless, iodine - purple)
Conductivity	Good conductors of heat and electricity	Poor conductors of heat and electricity (e.g., oxygen, sulfur)
Ductility	Ductile (e.g., copper, gold)	Not ductile
Malleability	Malleable (e.g., gold, aluminum)	Not malleable (e.g., sulfur)
Sonorous	Sonorous (e.g., copper, brass)	Non-sonorous



" Burning a candle involves both physical (melting wax) and chemical changes (wax burning)."

Some elements have intermediate properties between those of metals and nonmetals, they are called **metalloids**; examples are boron, silicon, germanium, etc

Compounds : A compound is a substance composed of two or more elements, chemically combined with one another in a fixed proportion.

Mixtures	Compounds
1. Elements or compounds just mix together to form a mixture and no new compound is formed.	1. Elements react to form new compounds.
2. A mixture has a variable composition.	2. The composition of each new substance is always fixed.
3. A mixture shows the properties of the constituent substances.	3. The new substance has totally different properties.
4. The constituents can be separated fairly easily by physical methods.	4. The constituents can be separated only by chemical or electrochemical reactions.